

A Short Demo of $\text{\LaTeX}$.js

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Abstract

This page is rendered live in your browser by $\text{\LaTeX}$.js, a $\text{\LaTeX}$ to HTML5 translator written in JavaScript. It shows a small but complete document: sections, lists, tables, figures, mathematics and footnotes.

1 Introduction

$\text{\LaTeX}$.js parses $\text{\LaTeX}$ source and produces HTML together with a stylesheet that mimics the default $\text{\LaTeX}$ output. Unlike a math-only renderer such as KaTeX or MathJax, it understands the document *structure* as well: sections, lists, tables, figures and cross references.

The idea is simple. You write ordinary $\text{\LaTeX}$, and the result is a normal web page — no $\text{\TeX}$ installation, no build step on the server.

1.1 What this document covers

- sectioning: sections and subsections
- lists: both unordered and ordered
- a table with a caption
- a figure with an image and a caption
- mathematics: inline, numbered equations and aligned equations
- typography and footnotes

2 Lists

An ordered list behaves as expected:

1. Parse the $\text{\LaTeX}$ source.
2. Build the document tree.
3. Emit HTML and apply the stylesheet.

Table 1: Three projects that translate $\text{\LaTeX}$ for the web.

Project	Kind	Bundle size
$\text{\LaTeX}$.js	full document	593 KB
KaTeX	math only	280 KB
MathJax	math only	1.1 MB

[width=8cm]/img/stars.png

Figure 1: A figure placed with its own caption.

3 Tables

4 Mathematics

Inline mathematics sits in the flow of a sentence: for instance $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ is the familiar closed form. A numbered equation is set on its own line and labelled on the right:

$$\int_0^1 x^2 dx = \frac{1}{3} \tag{1}$$

Several lines can be aligned on a common symbol:

$$\begin{aligned} (a + b)^2 &= a^2 + 2ab + b^2 \\ (a - b)^2 &= a^2 - 2ab + b^2 \end{aligned}$$

Subscripts and superscripts compose naturally, as in $a^2 + b^2 = c^2$, and Greek letters such as α , β and γ are available.

5 Figures

Images are included with `\includegraphics` and floated with a caption. The width can be given in any familiar unit:

6 Typography

The Computer Modern fonts are shipped with the renderer, so the output really does look like a $\text{\LaTeX}$ document rather than a web page pretending to be one¹. Ligatures, spacing and the shape of the letters all come from the same typefaces $\text{\TeX}$ uses.

¹This is a footnote. In print it would sit at the bottom of the page; on the web it is collected at the end of the document.

7 Conclusion

Everything on this page was produced from plain $\text{\LaTeX}$ source, inside the browser, with no server-side processing at all.